

Automated Reddit Data Pipeline for Airline Sentiment Analysis Using Prefect and AWS S3

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Abstract—Social media platforms such as Reddit offer valuable perspectives on public sentiment and customer experiences concerning airline services. This study presents an automated data pipeline developed to gather, process, and store Reddit discussions referencing prominent airlines like Emirates, Lufthansa, and Cathay ,KLM, Delta, Qatar etc. The system employs the Prefect orchestration tool to streamline the Extract–Transform–Load (ETL) workflow, enabling seamless data extraction via the Reddit API (PRAW) and long-term storage on Amazon S3. Through deployment and scheduling on Prefect Cloud, the pipeline performs periodic data retrieval, supporting near real-time sentiment monitoring. This architecture ensures scalable and continuous data acquisition, forming a reliable foundation for analyzing customer satisfaction trends and predicting sentiment patterns across the airline industry.

Index Terms—Reddit Data Extraction, Airline Sentiment Analysis, Prefect, AWS S3, ETL Pipeline, Automation, Social Media Analytics

I. INTRODUCTION

Social media platforms are becoming essential outlets for people to communicate their opinions, experiences, and criticism on different services in the digital era. One of the biggest discussion-based websites, Reddit, has a huge volume of user-generated content that represents public opinion and real-world attitudes in a variety of businesses. In the aviation industry, conversations on travel experiences, airline performance, and customer service offer a useful tool for assessing customer satisfaction and pinpointing areas for service improvement.

Such online conversation may be used by airlines and researchers to track company reputation, evaluate consumer complaints, and anticipate new trends in passenger expectations. However, Reddit’s massive volume, unstructured nature, and ongoing data production make it extremely difficult to manually collect and analyse this data. Automation and cloud-based data pipelines provide effective, scalable, and real-time data acquisition and administration options to solve these issues.

An automated Reddit data pipeline for gathering, processing, and storing airline-related discussions for sentiment analysis is presented in this study. The suggested solution automates the Extract-Transform-Load (ETL) process by integrating the Reddit API with the Prefect orchestration framework. In order to provide a centralised data lake for additional analysis and visualisation, collected data is methodically stored on Amazon Simple Storage Service (AWS S3). Prefect Cloud is used to schedule the pipeline, allowing for automated periodic execution. In addition to streamlining data engineering processes,

this method creates a solid basis for sentiment analytics, allowing for ongoing public sentiment monitoring of major airlines like Emirates, Lufthansa, Cathay, and others.

II. LITERATURE REVIEW

[1] In modern data engineering, Apache Airflow is one of the most popular open-source workflow orchestration technologies [1]. Their study highlights Airflow’s Directed Acyclic Graph (DAG) design, which enables efficient scheduling, dependency management, and monitoring of difficult ETL processes. The framework’s integration with Kafka for real-time data streaming and support for cloud-based deployments guarantees scalability and reliability in distant scenarios. With Airflow’s versatility, organisations may automate end-to-end data pipelines, simplifying work-flows for data intake, transformation, and analysis. It is therefore an essential tool for processing massive amounts of data where speed and fault tolerance are crucial. [2] demonstrates the practical implementation of Airflow’s built-in operators, task dependencies, and parallel execution capabilities on Google Cloud Platform (GCP). The study demonstrates how Airflow’s orchestration capabilities simplify ETL processes, improve workflow performance and scalability, and enable data preparation for analytics and machine learning pipelines. All things considered, these initiatives make Apache Airflow a key orchestration framework for managing complex, automated, and cloud-integrated data engineering procedures.

[3] A crucial component of modern cloud infrastructure distributed storage systems is Amazon Simple Storage Service (S3). Their study shows how automatic replication across several sites ensures consistent data availability and integrity, hence achieving S3’s fault tolerance and longevity. It is emphasised how S3’s smooth interaction with analytics and data processing systems allows for reliable, large-scale data storage options suitable for data-intensive applications. By using its distributed and scalable architecture, S3 serves as the cornerstone for cloud-based storage, enabling a range of data-driven and business analytics applications.

[4]examines Amazon Redshift, a data warehousing solution that significantly depends on S3 for backup and long-term storage. Their research describes Redshift’s massively parallel processing (MPP) capabilities and columnar storage architecture, which allow for high-performance querying over petabyte-scale datasets. The authors point out that Redshift’s

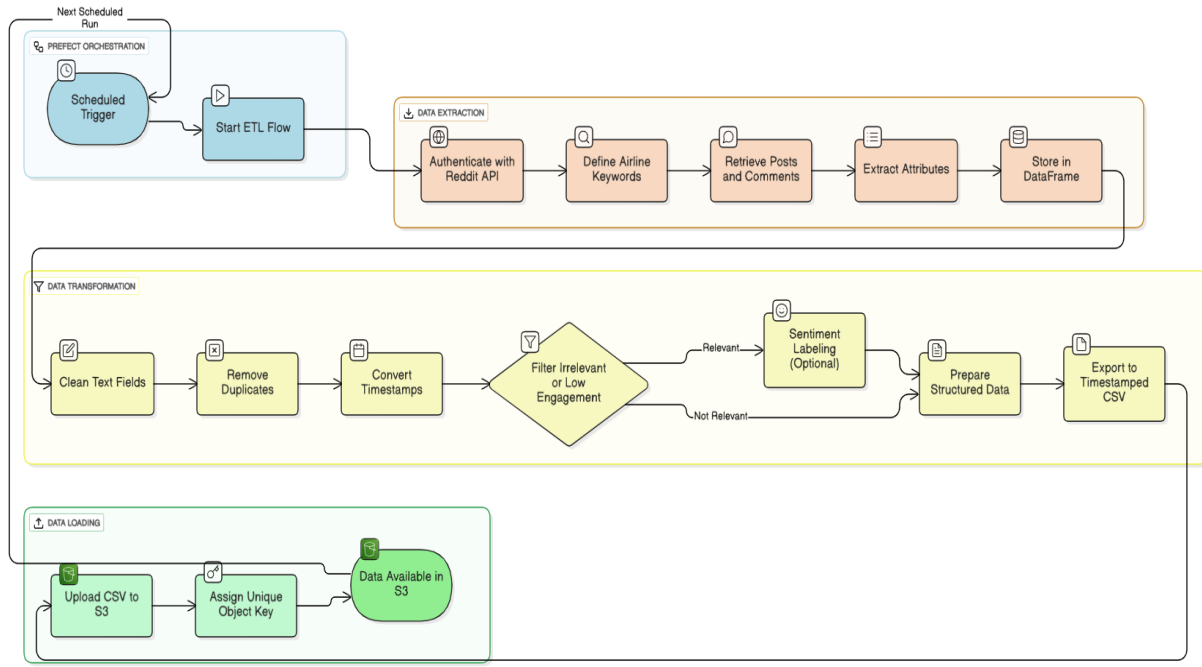


Fig. 1. Architecture of the Reddit-to-S3 ETL Pipeline with Prefect Orchestration

analytical efficiency is based on S3’s dependability and flexibility, allowing for quick, affordable, and scalable analytics in enterprise data ecosystems. When taken as a whole, these studies establish Amazon S3 as a crucial facilitator of distributed data storage and analytics, offering the fundamental foundation for high-throughput, scalable, and reliable data systems in cloud settings.

[5]The need of effective data orchestration, governance, and analytics in handling large-scale and heterogeneous datasets has been highlighted by recent developments in data engineering. developed an integrated framework to enable scalable data intake, transformation, and analytical processing in distributed contexts by integrating tools like Apache Spark, Apache Airflow, and Kafka. This design demonstrates how orchestration layers may efficiently ensure data quality, consistency, and compliance while streamlining batch and streaming data operations. Through the use of cloud-native architectures that make use of data lakes—like Amazon S3—and automated workflow orchestration systems for real-time data analytics, further research has expanded these concepts. However, current enterprise-level orchestration solutions, such as Airflow, sometimes need substantial computing resources and have steep learning curves, making them difficult for small-scale and academic applications. Lightweight workflow orchestrators like Prefect, which provide improved usability, streamlined scheduling, and simple dependency management, have emerged as viable solutions to lessen these restrictions. Data pipeline automation and monitoring are made possible by Prefect’s smooth interaction with cloud platforms and APIs, such as Reddit and AWS S3. This development is part of a larger trend towards cloud-based, accessible, resource-

efficient orchestration frameworks that support reproducibility, scalability, and transparency in data-driven research settings.

III. METHODOLOGY

Reddit data pertaining to airline talks is gathered, processed, and stored by the project using an end-to-end Extract–Transform–Load (ETL) [6] pipeline (Fig. 1). Open-source Python libraries like [7] for data extraction, pandas for transformation, and AWS S3 for cloud storage are used in the workflow’s implementation. Each csv file extracts 200 reddit posts.

Prefect, which automates task execution, scheduling, and monitoring, is used to coordinate the whole process. Because Prefect offers a simpler setup, Python-native syntax, and a lightweight structure appropriate for students and research projects, it was selected over Apache Airflow. It doesn’t need any additional infrastructure, such as web servers or schedulers, and may be executed straight from scripts.

The project is entirely affordable and available for academic exploration thanks to the usage of all platforms and resources, including the AWS Free Tier, Prefect’s free community edition, and Reddit’s public API. [8]

A. Data Extraction

Reddit posts and comments mentioning well-known airlines like *Emirates*, *Lufthansa*, *Qatar Airways*, and *Cathay Pacific* are retrieved during the data extraction phase. Reddit’s public API may be accessed and authenticated using the Python Reddit API Wrapper (PRAW). Airline-related keywords are used to create queries, and pertinent information is retrieved, including post titles, timestamps, upvotes, comment counts,

and subreddit names. For additional processing, the extracted data is momentarily stored in Pandas DataFrames. .

B. Data Transformation

The data is transformed after extraction to guarantee consistency and usefulness. Text fields must be cleaned, duplication must be eliminated, timestamps must be converted to standard formats, and irrelevant or low-engagement postings must be filtered. After that, the organised data is ready for export as a CSV file. For version control and repeatability, each pipeline execution creates a timestamped file (e.g., `reddit_airline_posts_20251015.csv`)

C. Data Loading

The [9]library is used to upload the processed CSV file to an Amazon Web Services (AWS) S3 bucket (fig. 2). S3 offers a scalable and dependable cloud storage option for storing massive amounts of gathered Reddit data. Every version of the dataset is kept under a distinct object key, enabling downstream analysis, comparison, and future retrieval. All team members and systems may safely access data thanks to its cloud-based architecture, which eliminates the need for local storage.

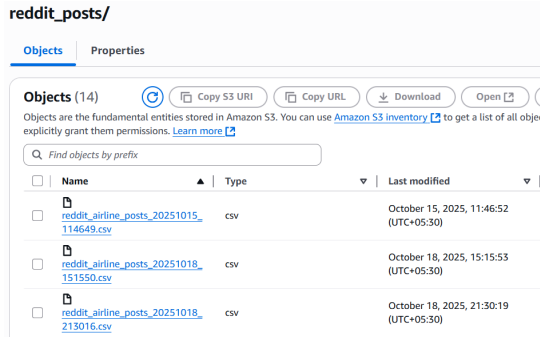


Fig. 2. aws-s3

D. Orchestration with Prefect

The Prefect [10]framework manages scheduling and process automation. The main flow is defined in the `reddit.py` file:

```
from prefect import flow
from reddit_to_s3 import reddit_to_s3_flow

if __name__ == "__main__":
    reddit_to_s3_flow.serve(
        name="reddit-flow",
        cron="0 */10 * * *"
    )
    # runs 10 every hours
```

The `reddit_to_s3_flow` is deployed by this code as a Prefect flow called `reddit-flow`, which is set up to run automatically every ten hours depending on the supplied CRON expression. With Prefect Cloud's real-time task status visibility, customers can easily manage retries, reschedule flows, and keep an eye on runs.

IV. VISUALIZATION AND ANALYSIS

A straightforward Python script was used to analyse individual CSV files in order to extract and visualise important insights for analysis and visualisation. Sentiment distribution, airline mentions, and the frequency of positive and negative terms were the three primary areas of attention for the analysis. A Python-based machine learning script was used to analyse the CSV files in order to calculate counts, carry out sentiment categorisation, and display patterns in the data..

To illustrate the distribution of airline mentions across Reddit posts, a bar chart (fig. 3) was created. United Airlines and Qatar Airways received the highest number of references, each approaching 30 mentions. American Airlines also had a substantial presence, followed by Delta and Singapore Airlines with fewer mentions. In contrast, airlines such as Lufthansa, Emirates, KLM, Air France, and Cathay Pacific had no recorded mentions, highlighting a significant disparity in online discussion volume.

The distribution of attitudes and the general balance between positive and negative feedback, a bar chart (fig. 4) was created. American Airlines received the highest number of sentiment mentions, with 7 categorized as "Good" and 12 as "Bad." Delta Airlines had only positive mentions (3), while United Airlines had only negative mentions (3). Other airlines, including Lufthansa, Emirates, KLM, Air France, Singapore, and Cathay Pacific, had no recorded sentiment mentions in this chart.

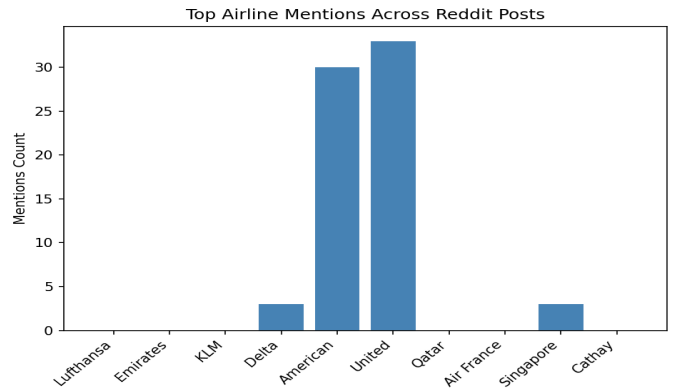


Fig. 3. Top Airline Mentions Across Reddit Posts

A. Why Prefect Over Apache Airflow

Prefect was selected for this project because of its lightweight setup, Pythonic architecture, and user-friendly interface, even though Apache Airflow is an industry-standard orchestration solution. Prefect is perfect for academic and research-focused projects since it removes the need to manage external schedulers and configure DAG files. Developers may create and test flows locally while retaining complete cloud-based visibility thanks to its smooth interface with Python functions. This improved Prefect's suitability and accessibility for a student research pipeline such as this one.

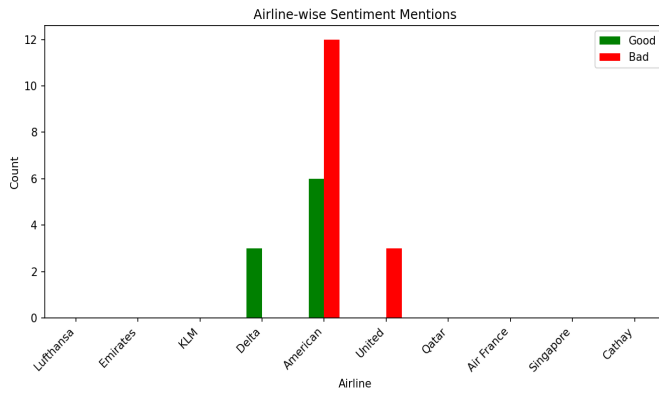


Fig. 4. Airline Sentiment Distribution (Good/Bad per Airline)

V. FUTURE WORK

This project can be enhanced in the future by utilising more efficient and scalable data processing and storage technologies. A major improvement would be switching to a commercial version of Amazon S3, which offers increased availability, faster retrieval times, and better security for managing large datasets. Workflow orchestration systems like Prefect may further automate and optimise data pipeline execution in comparison to manual or script-based scheduling, ensuring better reliability and monitoring. Analytically, the project may be expanded by putting in place a more reliable machine learning model that can manage bigger, more complicated datasets. More sophisticated methods like topic modelling or transformer-based sentiment models may offer more in-depth understanding of user comments. When combined, these enhancements would increase the system's scalability, accuracy, and suitability for enterprise-level data analysis and decision-making.

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